Intermittent working-memory coding through NMDA-sustained persistent activity

Joao Barbosa

Cognitive Neuroimaging Unit, INSERM, CEA, CNRS, Université Paris-Saclay, NeuroSpin center & Institut de neuromodulation, GHU Paris, psychiatrie et neurosciences, centre hospitalier Sainte-Anne, pôle hospitalo-universitaire 15, Université Paris Cité, Paris, France.

Abstract

Working memory is classically attributed to persistent spiking sustained by slow NMDA receptor currents. This view has been challenged by large-scale prefrontal recordings in which the remembered item is decoded only intermittently on single trials, suggesting that memories survive silent periods through synaptic facilitation. Here we show that intermittent coding emerges from canonical attractor networks with fixed synapses and a fluctuating, attention-like global input. Brief drops in attention silence the network, but NMDA currents can bridge each silence. Fitted to monkey prefrontal recordings, a rate model reproduces the number and duration of silent states and predicts how they change with the decoded neurons. We found that silent states are more frequent in less informative samples, validating this prediction and suggesting that measured silences overestimate how often memory leaves spiking activity. Unlike synaptic facilitation, the NMDA trace also predicts memory loss when silences outlast a few hundred milliseconds. In line with this, monkeys made more errors when silent states were longer and their performance pointed to a trace of ∼120 ms, matching NMDA kinetics rather than the long time scale of facilitation. Finally, we replicate the main findings in a biophysical ring-attractor network of spiking neurons with NMDA-mediated recurrent excitation, demonstrating intermittent rate coding through persistent activity bridged by NMDA currents.

Introduction

Two main mechanisms have been proposed to hold information in working memory. In persistent-activity models, recurrent excitation keeps a cue-selective population firing through the delay, and slow NMDA receptor currents make this activity stable (Amit and Brunel, 1997; Wang, 1999, 2001; Compte et al., 2000). Consistent with these models, blocking NMDA receptors markedly reduces persistent firing in monkey prefrontal cortex (Wang et al., 2013). In activity-silent models, the memory is instead stored in short-term synaptic facilitation and does not require ongoing spiking during maintenance (Mongillo et al., 2008; Stokes, 2015; Lundqvist et al., 2016; Miller et al., 2018; Panichello et al., 2024), and hidden memories can be reactivated after periods without decodable activity (Rose et al., 2016; Wolff et al., 2017). Notably, the two accounts rely on traces with very different time scales. NMDA receptor currents decay on a time scale of hundreds of milliseconds (Vicini et al., 1998; Wang et al., 2008), and models of persistent activity typically use an NMDA decay time constant of about 100 ms (Wang, 1999; Compte et al., 2000). On the other hand, facilitation at excitatory synapses in prefrontal cortex decays on a time scale of the order of a second (Thomson, 1997; Markram et al., 1998; Hempel et al., 2000; Wang et al., 2006; Mongillo et al., 2008), and activity-silent models typically use 1.5 s (Mongillo et al., 2008). Synaptic augmentation, which lasts tens of seconds, has been proposed to bridge even longer silences (Mongillo and Tsodyks, 2025).

Rigorously testing predictions of these models requires single-trial resolution (Lundqvist et al., 2018), which only recently became possible with the advance of large-scale, intracranial recordings

(Williams and Linderman, 2021; Urai et al., 2022). Recent studies show that working-memory information in prefrontal cortex (PFC) fluctuates within a trial (Panichello et al., 2024; Tao and Libedinsky, 2024; Mozumder et al., 2026). Panichello et al. (2024) found that decoding alternates between *On states* (when the memory can be read out) and *Off states* (when it cannot) and concluded that synaptic changes carry the memory through *Off states*. Tao and Libedinsky (2024) found rare periods of 43–414 ms in which all simultaneously recorded selective neurons were silent for longer than expected by chance, and concluded that short-term plasticity is needed in a small fraction of trials. Finally, Mozumder et al. (2026) found that persistent activity is asynchronous across neurons, with no population silences beyond chance. In their recordings, decoding *Off states* came with reduced but not abolished tuning and were synchronized between prefrontal and parietal cortex, and the authors interpreted them as global fluctuations that exert some influence on behavior.

Here, we ask whether a single persistent-activity mechanism can account for these observations and propose that intermittent decoding does not require short-term plasticity and is in fact more consistent with the NMDA account. To illustrate this, we modelled working memory with a classic attractor model, in which two cue-selective populations sustain their firing through recurrent excitation mediated by slow NMDA currents and compete through shared inhibition (Wang, 2002; Wong and Wang, 2006). The synchrony of *Off states* seen across prefrontal and parietal cortex (Mozumder et al., 2026) points to a common input as their source, and attention, which waxes and wanes on sub-second time scales (Fiebelkorn and Kastner, 2019), is a natural candidate. We therefore added to the model a fluctuating attentional input common to both populations, whose brief drops silence both populations. During each silence slow NMDA currents carry the memory across, so the cued population resumes firing when the attentional signal returns. We read out the model with the same decoding and *On/Off* criteria as Panichello et al. (2024) and fit how often, for how long and how deeply attention lapses to the *On/Off* statistics of these recordings. The model quantitatively reproduces the number and length of *On and Off states* while keeping the memory in persistent activity. Our reanalyses of the PFC recordings showed that *Off states* are more frequent when fewer (or less selective) neurons are decoded, so their frequency partly reflects how well the recorded sample reads out the memory, and the attractor model predicts this dependence for decoded populations it was never fitted to. Unlike a model in which short-term facilitation carries the memory, the NMDA model predicts that errors grow with the time spent *Off*, as we found in the recordings. Fitted to the monkeys' performance, the decline in accuracy with *Off* time points to a trace of about 120 ms, which matches NMDA kinetics rather than facilitation. Finally, we leverage the insights from the rate model to build a ring-attractor network of spiking neurons modulated by a global signal and replicate the main results of Panichello et al. (2024). Previous work took intermittent decoding as evidence that persistent activity is not sufficient for working memory, our results show that this evidence has a more likely explanation within the standard persistent-activity account.

Results

Input dips produce *Off states* and NMDA carries the memory across them

We started by modelling *On/Off states* using a variation of the classic two-pool attractor model (Wang, 2002; Wong and Wang, 2006), with two memory pools, cued and uncued, that excite themselves through NMDA currents and compete through shared inhibition (Methods). The key addition to the classic model is a common input to both pools that fluctuates, which we interpret as an attentional signal, since attention is known to wax and wane on sub-second time scales (Fiebelkorn and Kastner, 2019; see Discussion). Without fluctuations in its input, the cued pool stays active after the cue and decoding stays *On* (Fig. 1a, first second after the cue). With brief dips in this input, both pools fall silent and decoding drops to chance (Fig. 1a, after 1,000 ms). During each dip, the NMDA gating of the cued pool decays with a time constant of

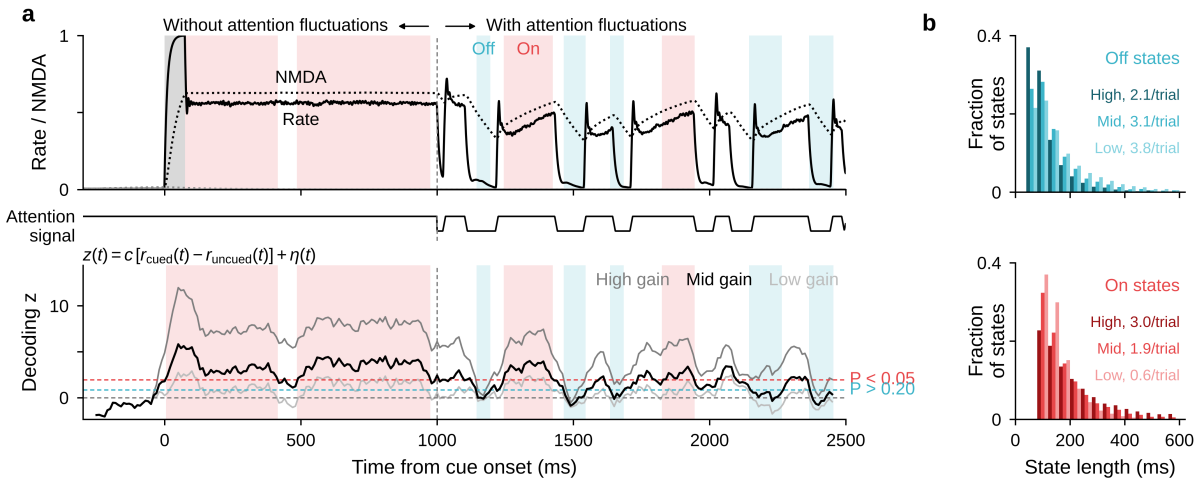


Figure 1: Input dips create *On and Off states*, and NMDA carries the memory. a, Example trial. Top: rates of the cued (black) and uncued (grey) pools, relative to the maximum rate, and their NMDA gating (dotted); grey shading marks the cue. Middle: common input (attention signal), constant for the first 1,000 ms after the cue and with dips afterwards. The dip-free first second is shown for illustration only; in all other simulated trials, dips occur throughout the delay. Bottom: decoding z score of the same trial read out with the calibrated gain c (mid gain, black), twice this gain (high gain, dark grey) and half of it (low gain, light grey), with the same read-out noise. *On* (red) and *Off* (cyan) states are those of the mid-gain read-out. Dashed lines: *On* threshold (z = 1.96) and *Off* threshold (z = 0.84), as in Panichello et al. (2024). **b**, Length distributions of *Off* (top) and *On* (bottom) states in the model at the three gains (dark, high; medium, mid; light, low; 2,112 simulated trials), as fractions of states in 40 ms bins, with the mean number of states per trial.

150 ms and stays higher than that of the uncued pool. When the input returns, the pool with more NMDA gating wins the competition. *Off states* therefore do not erase the memory, because NMDA gating carries it across each silence, and the cue is still correctly remembered at the end of the delay.

To compare the model with the data, we read it out like the recordings (Supplementary Fig. 1; Methods). In the data, a decoder trained on the recorded neurons gives every 10 ms a z score of how well the cue can be read out. The model has no neurons, so we built the equivalent score from its two pools. The evidence for the cue is the difference between the cued and uncued pool rates (Δr) in the same 100 ms windows. We converted this difference into z units by multiplying it by a gain c and adding read-out noise η, whose variance and temporal correlations were estimated from PFC recordings using the decoder output before the cue (Methods). We confirmed the validity of this method by showing that when synthetic neurons driven by the model were decoded with the same pipeline, the decoder's z agreed with this read-out as closely as two runs of the decoder on independent datasets (Supplementary Fig. 1e–i). *On and Off states* (red and cyan in Fig. 1a) are then labelled with the thresholds of Panichello et al. (2024) (dashed lines in Fig. 1a). Read out this way, the model alternates between *On and Off states* (Fig. 1a, bottom, mid gain). Both length distributions peak near 100 ms and have long tails (Fig. 1b), qualitatively like those in PFC recordings (Panichello et al., 2024). Importantly, the *On/Off* statistics depend strongly on the decoder gain. Keeping the dynamics and noise fixed for this particular simulated trial, halving the gain gave more *Off states* and almost no *On states*, whereas doubling it gave fewer and shorter *Off states* and more *On states* (Fig. 1a, bottom, low and high gain, and Fig. 1b for many trials). In the next sections, we test this prediction and ask whether the model matches the data quantitatively.

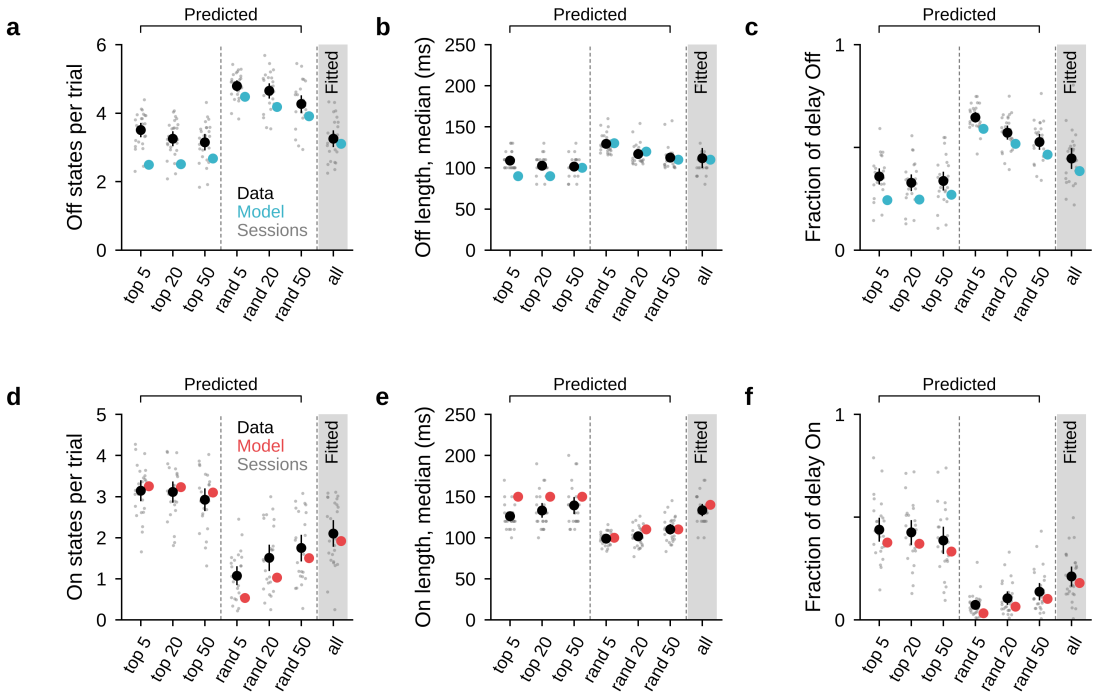


Figure 2: The model predicts how *On/Off* statistics change with the decoded population. Decoders using the most cue-selective neurons (top 5, 20, 50) are *On* more often and *Off* less often than decoders using the same numbers of random neurons (rand 5, 20, 50). Black circles: data, mean ± 95% CI (2 SEM) over 25 sessions. Grey dots: single sessions. Cyan and red circles: model, *Off and On* statistics. The model was fitted to all neurons only (grey band). The other six decoded populations are predictions obtained by changing only the read-out gain and noise (Methods).

On/Off statistics depend on which neurons are decoded

In our reanalysis of the PFC recordings of Panichello et al. (2024), we found that *On/Off* statistics depend strongly on which neurons the decoder uses, which we call the decoded population (Fig. 2, black circles). Decoders using the most cue-selective neurons spend much more of the delay *On*, and less *Off*, than decoders using the same numbers of randomly chosen neurons, which also show more *Off states* per trial (paired Wilcoxon signed-rank tests over 25 sessions, $p < 10^{-6}$ for each of 5, 20 and 50 neurons; Fig. 2). How often a memory appears *Off* therefore depends partly on how well the recorded neurons read it out. This matters for interpreting *Off states* as evidence for activity-silent memory. A decoder trained on a limited sample of neurons can fall to chance while the memory is still present in the activity of other, potentially unrecorded neurons. The measured fraction of time *Off* is therefore an upper bound on how often the memory leaves spiking activity. We then asked if our model could explain this dependence.

To test this, we fitted the model dynamics to the decoder using all neurons in 25 sessions (Methods). The model matches the data well in *Off states* per trial (3.1 vs 3.3), *Off* median length (110 vs 112 ms), *On states* per trial (1.9 vs 2.1) and *On* median length (140 vs 134 ms) (Fig. 2, grey band). We then kept the dynamics fixed and, for each of the six other decoded populations, changed only the read-out without refitting any parameters. Its gain was calibrated on that population's decoding strength and its noise on that population's pre-cue z (Methods). The model quantitatively predicts the shift between selective and random neurons (Fig. 2; Supplementary Fig. 2).

Errors increase with time *Off*, as expected from an NMDA trace

So far, we analysed correct trials only, but error trials can help distinguish between the two accounts of working memory. NMDA currents and short-term synaptic facilitation can both carry a memory across brief *Off states*, but on very different time scales. In our model, NMDA gating decays with a time constant of 150 ms, whereas facilitation decays on a time scale of the order of a second (Wang et al., 2006; Mongillo et al., 2008), comparable to the length of the whole memory period. To see explicitly what each mechanism predicts regarding error trials, we compared our model with a version in which facilitation replaces NMDA gating (STP model; Methods). Fitted in the same way, it reproduced the *On/Off* statistics about as well as the NMDA model (Supplementary Fig. 3). This is expected, because in both models *Off states* are driven by lapses of attention, which facilitation (τf = 1.5 s) bridges even more easily than NMDA gating (τ_N = 150 ms). We then silenced each model with a single dip in the common input of increasing duration (Methods). The NMDA model held the memory for silences of up to 350 ms and lost it for every silence from 400 ms on, because the NMDA trace had decayed too far for the cued pool to restart (about 2.5 τ_N), whereas the STP model held the memory for silences as long as 1 s (Fig. 3a). The NMDA account therefore predicts that errors become more frequent when the memory stays *Off* for long, whereas facilitation predicts no such relation for silences shorter than its own decay, which here spans the whole delay.

We tested this prediction by comparing *On/Off* statistics between correct and error trials (Methods). Within sessions, the odds of an error increased with the total *Off* time in the delay and with the mean length of *Off states*, and decreased with the total *On* time, but did not depend on the number of *Off states* (Fig. 3b). Errors thus depended on how long the memory was *Off*, not on how often it went *Off*. Frequent *Off states* are harmless as long as each is brief. Across all 25 sessions, total *Off* time increased the odds of an error (β = 0.32, 95% CI 0.20–0.43, $p = 1\times10^{-5}$; logistic mixed model with sessions as random effects, Methods). The effect was clear in monkeys H and J (β = 0.48, $p = 1\times10^{-4}$, and β = 0.52, p = 0.005) and weaker in monkey A (β = 0.10, p = 0.13).

Finally, we asked which time constant of the trace best explains how performance falls with *Off* time. We varied τ_N in the fitted NMDA model, keeping the dip statistics and all other parameters unchanged, and simulated each version with the same random dips, read out as the data. We linked the probability that the memory was held to the probability of a correct response, assuming that the monkeys guess among the 8 locations after a memory loss (Methods). With the best-fitting τ_N of 123 ms, the model reproduced the decline of the pooled fraction correct (Fig. 3c). Resampling sessions gave a 95% confidence interval of 101–190 ms (Fig. 3d). The decline of performance with *Off* time thus calls for a trace of about 120 ms, close to measured NMDA receptor kinetics and to the 150 ms assumed in the fitted model, and far shorter than the time scale of facilitation.

Intermittent rate coding in a biophysical ring-attractor network without short-term plasticity

The rate model describes the activity of two populations, with one discrete attractor for each memory. It leaves open whether its mechanism works in a continuous attractor, which can hold any location and along which the memory could drift during silences, and in a network of spiking neurons, where NMDA currents, recurrent inhibition and noise interact at the level of single cells. We therefore turned to the spiking network of Compte et al. (2000), a ring of 2,048 excitatory and 512 inhibitory integrate-and-fire neurons in which recurrent excitation is mediated by NMDA receptors and a localized bump of persistent activity holds the cued location. We used this network with minimal changes (Methods). The NMDA decay time was set to 150 ms, as in the rate model, the neurons were made heterogeneous, and all excitatory neurons received the same kind of fluctuating attentional input as the rate model. The network was not fitted to the data.

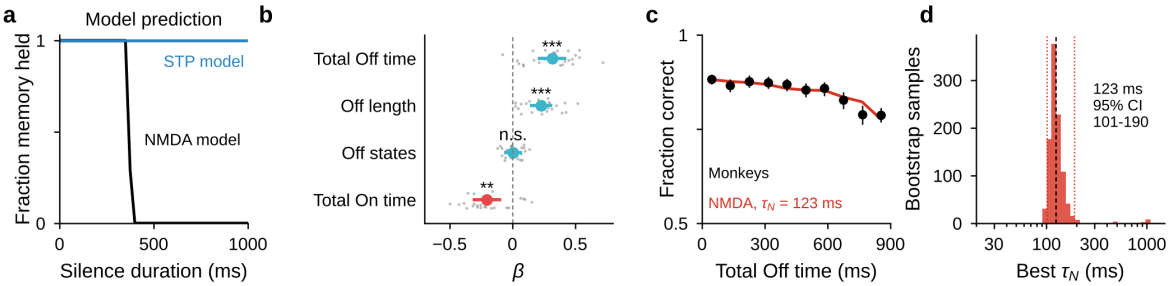


Figure 3: Errors increase with time *Off*, and the decline in performance points to a trace of about 120 ms. a, Model prediction (single-dip test): fraction of trials with the memory held against the duration of a single silence, imposed by one dip of the common input (0 to 1 s in 25 ms steps, 300 trials per duration), for the NMDA model (black) and the STP model (blue). Memory was scored on the model, without read-out (Methods). **b**, Effect on the log-odds of an error (β) per within-session SD of four statistics of the delay (500–1,400 ms after cue onset): total *Off* time, mean *Off-state* length, number of *Off states* and total *On* time. Coloured circles and lines: fixed effect and 95% CI of a logistic mixed model, fitted for each statistic to all 25 sessions (Methods). Grey dots: single sessions (fixed effect plus the session's random slope). Likelihood-ratio tests: total *Off* time, $p = 1{\times}10^{-5}$; *Off* length, $p = 7{\times}10^{-5}$; number of *Off states*, $p = 0.89$; total *On* time, $p = 0.001$. $^{***}p < 0.001$, $^{**}p < 0.01$, n.s. not significant. **c**, Fraction correct against total *Off* time in the delay, all 15,468 trials pooled, in 90 ms bins (black; 95% Wilson interval), and the NMDA model with the best-fitting τ_N of 123 ms (red), linked to behaviour by P(correct) = p0 × [P(held) + P(lost)/8] (Methods). **d**, Best τ_N in 1,000 bootstraps over sessions. Dashed: fit to all data. Dotted: 95% interval (101–190 ms).

We sampled 330 of its neurons, the average number of units recorded per session, and read them out with the same decoder, null and *On/Off* labelling as the recordings.

With a constant attentional input, the cued location was held by a stable bump of persistent activity, and decoding stayed *On* throughout the delay (Fig. 4a, first second after the cue). Each drop of the input silenced the bump, and decoding fell to chance. When the input returned, the bump restarted at the cued location, because the cued neurons kept more NMDA gating through the silence than the other neurons, as in the rate model. Read out like the recordings, the network therefore alternated between *On and Off states* (Fig. 4a, bottom). Both state lengths had long-tailed distributions, as in the rate model (Fig. 4b).

Finally, Panichello et al. (2024) found that neurons lose their memory tuning during *Off states* and concluded that synaptic changes carry the memory through these periods. We applied their cross-validated analysis to the spiking network. Held-out neurons were strongly tuned to the cued location during *On states* and barely tuned during *Off states* (Fig. 4c), as in the recordings. In the plane of the eight cue-condition means, delay activity of both the network and the recordings moved toward the centre during *Off states* (Fig. 4d), and its alignment with the cued location dropped but remained above its level before the cue (Fig. 4e). Even when the decoder labelled the memory *Off*, population activity therefore still held a weak code for the cued location. *Off states* defined by a decoder thus overestimate how often the memory leaves spiking activity, as the dependence of *On/Off* statistics on the decoded population also suggests (Fig. 2). Together, these results show that intermittent rate coding, including the loss of tuning in *Off states*, emerges in a biophysical attractor network without short-term plasticity.

Discussion

Recent large-scale recordings found that prefrontal decoding of a remembered item is intermittent on single trials (Panichello et al., 2024; Tao and Libedinsky, 2024; Mozumder et al., 2026), suggesting that memories survive silent periods through synaptic facilitation (Mongillo et al.,

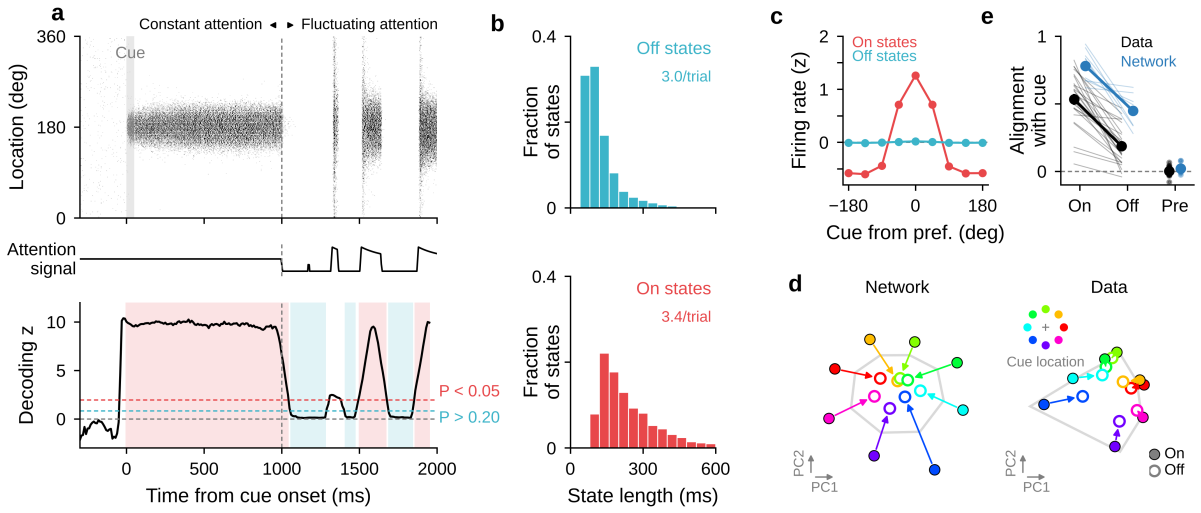


Figure 4: Intermittent rate coding in a spiking ring-attractor network with NMDA currents. **a**, Example trial. Top: spikes of the 2,048 excitatory neurons, sorted by preferred location (counts in 5 ms and 8-neuron bins); grey shading marks the cue (180°). Middle: common input (attention signal) to the excitatory neurons, constant for the first 1,000 ms after the cue and with dips afterwards, each followed by a transient overshoot. The dip-free first second is shown for illustration only; in all other simulated trials, dips occur throughout the delay. Bottom: decoding z score from 330 simulated neurons, with the decoder, null and *On* (red) and *Off* (cyan) labelling of the recordings. Dashed lines: *On* threshold ($z = 1.96$) and *Off* threshold ($z = 0.84$). **b**, Length distributions of *Off* (top) and *On* (bottom) states, with the mean number of states per trial (4 simulated sessions of 480 trials). States longer than 600 ms not shown. **c**, Cross-validated memory tuning of held-out simulated neurons during *On* (red) and *Off* (cyan) states, computed as in Panichello et al. (2024) (mean ± s.e.m., 330 neurons). **d**, Delay activity in the plane of the first two principal components of the eight cue-condition means, for one simulated session (left) and one recorded session (221024, right). Grey, the cue-condition means. Filled and open circles, the mean position of each cue's trials during *On and Off* moments, with an arrow from *On* to *Off*. Colours, cue location (inset). **e**, Alignment of each trial's position with its cue's direction (cosine, minus the same with shuffled cue labels) during *On and Off* moments and before the cue (Pre). Thin lines, sessions. Thick, mean ± s.e.m. (data, 25 sessions, black; network, 10 sessions, blue). Alignment above Pre during *Off* moments shows that population activity still coded the cue while decoding was *Off*.

2008; Stokes, 2015; Lundqvist et al., 2016; Miller et al., 2018; Panichello et al., 2024). We showed that the same *On/Off* statistics arise in conventional attractor models with fixed synapses. In the models, *Off states* occur when a common input briefly drops, and the memory survives each silence because NMDA receptor currents decay slowly enough to bias which pool restarts. Fitted to the decoder trained on the empirical data, the rate model reproduced the number and length of *On and Off states* while keeping the memory in persistent activity. Unlike a model in which short-term facilitation carries the memory, it also predicted that errors grow with the time spent *Off*, which we validated in prefrontal recordings.

In the recordings, *Off states* were more frequent and *On states* rarer when the decoder used random rather than selective neurons. This showed that a decoder falls to chance whenever the memory signal in its sample is weak relative to its noise, not only when the memory is absent from the network. The fraction of time *Off* measured from a limited sample is therefore an upper bound on how often the memory leaves spiking activity, and comparisons of *Off-state* statistics across studies or brain areas need to take the decoded population into account. The model captured this shift without refitting by changing only the read-out gain and noise, both calibrated from the data.

Our account is consistent with the other recent recordings of *Off states*. Mozumder et al.

(2026) found that *Off states* were synchronized between prefrontal and posterior parietal cortex, came with lower firing rates, and showed no obvious rhythmicity. A common input that reaches several areas and fluctuates irregularly, as in our fit, would produce such coordinated, aperiodic *Off states*. Tao and Libedinsky (2024) likewise found that information is reduced but present between gamma bursts. Our model is compatible with both, because NMDA currents can carry the memory across full silences but are not needed when activity is only reduced. While attention is sustained, the memory is held by persistent activity without any *Off state* (Fig. 1a, first second after the cue).

We interpreted the common input as an attentional signal that is high when attention is deployed on the memorandum and drops during lapses. Two lines of evidence support this interpretation. First, Engel et al. (2016) found that spiking in macaque area V4 alternates between *On and Off* episodes that occur synchronously across cortical layers. Importantly, these states were identified directly from population spiking, not from a decoder. *Off* episodes lasted about 100 ms on average (102 ms during stimulus-driven and 118 ms during spontaneous activity), close to the *Off states* in prefrontal decoding (about 110 ms) and within the time an NMDA trace can bridge. Spatial attention lengthened *On* episodes and made transitions less frequent, and monkeys detected a stimulus change more often when it occurred during an *On* episode than during an *Off* episode. Second, sustained attention samples the environment rhythmically rather than continuously. In humans, detection of threshold stimuli depends on the phase of ongoing EEG oscillations near 7 Hz (Busch et al., 2009), especially at attended locations (Busch and VanRullen, 2010), and when attention is divided between two locations, detection at each fluctuates at about 4 Hz (Landau and Fries, 2012). Similarly in monkeys, frontal eye field and lateral intraparietal activity alternates at theta frequencies (3–8 Hz) between periods of enhanced perception and periods in which attention is more likely to shift (Fiebelkorn et al., 2018), and human intracranial recordings show similar theta-rhythmic sampling in frontoparietal cortex (Helfrich et al., 2018). Because attention and working memory share prefrontal mechanisms (Awh and Jonides, 2001; Panichello and Buschman, 2021), lapses of attention could plausibly reach the memory network. In the fitted model, the attention signal recovers about five times per second, or one cycle every ∼200 ms. This is in the low theta range reported for attentional sampling (Landau and Fries, 2012; Fiebelkorn et al., 2018; Helfrich et al., 2018). Importantly, in contrast with these previous studies, the fitted signal was not periodic.

In our models, the memory is carried across each silence by NMDA gating that decays with a time constant of 150 ms and reflects recent firing rather than plasticity. This lies within the range of NMDA receptor currents, which decay over hundreds of milliseconds (Vicini et al., 1998; Wang et al., 2008) and are required for persistent firing in primate prefrontal cortex (Wang et al., 2013). *Off states* last about 100 ms, within the time an NMDA trace can bridge, so no longer-lasting mechanism is needed to explain them. The matching between behavioural simulations and data points also supports this interpretation. A memory held by facilitation decaying on a time scale of the order of a second should hardly be affected by a few hundred more milliseconds *Off*, as the STP model showed. We found that errors increased with the time spent *Off* and with the length of *Off states*, but not with their number (Fig. 3). This is expected from a trace that bridges brief silences but not long ones. Fitted to the monkeys' performance, the time constant of this trace was about 120 ms (95% CI 101–190 ms), close to measured NMDA receptor kinetics and far shorter than that of facilitation. Activity-silent mechanisms may still matter on longer time scales, for example to carry information between trials, when prefrontal memory traces are reactivated before the next stimulus (Barbosa et al., 2020). Notably, this between-trial trace is not a voluntary memory. It does not require attention and shows up as involuntary serial biases in behaviour, rather than as the information the subject actively maintains (Kiyonaga et al., 2017; Pascucci et al., 2023). Persistent activity and slower silent traces are therefore not mutually exclusive.

In sum, intermittent decoding of working memory is expected from an attractor network

with slow NMDA currents and a fluctuating, attention-like input. Its statistics depend on how well the recorded neurons read out the memory, and errors increase with the time spent *Off*, as expected from the limited duration of the NMDA trace. These results reconcile intermittent rate coding with persistent-activity accounts, identify a trace of about 120 ms that bridges *Off states*, and link *Off states* to the rhythmic but irregular nature of attentional sampling.

Methods

NMDA rate model

The model is the reduced two-pool attractor model of Wong and Wang (2006), with the changes described below. Two excitatory memory pools, $i \in \{\text{cued}, \text{uncued}\}$, have rates r_i and NMDA gating s_i. They share an inhibitory population r_I and receive a feed-forward inhibition q that follows the common input $m(t)$ with a lag. Parameters in bold were free during fitting; all others were fixed.

$$\tau_r \frac{dr_i}{dt} = -r_i + \varphi(x_i), \tag{1}$$

$$x_i = \boldsymbol{w_s} s_i + w_a r_i - w_I r_I + I_0\, m(t) - w_{ff}\, q + \text{cue}_i + \boldsymbol{\sigma}\xi_i, \tag{2}$$

$$\tau_N \frac{ds_i}{dt} = -s_i + \beta\, r_i(1 - s_i), \tag{3}$$

$$\tau_I \frac{dr_I}{dt} = -r_I + w_{EI}\,(r_{\text{cued}} + r_{\text{uncued}})\,, \tag{4}$$

$$\tau_{ff} \frac{dq}{dt} = -q + m(t). \tag{5}$$

The transfer function φ is a logistic sigmoid, $\varphi(x) = r_{\max}/(1 + \exp(-(x-1)/\kappa))$, with maximum rate $r_{\max}$, threshold 1 and slope κ, and ξ_i are independent Ornstein–Uhlenbeck noises ($\tau_\xi = 10$ ms). Fixed parameters: $r_{\max} = 150$, $\kappa = 0.06$, $w_a = 0.0141$, $\beta = 0.02$, $w_I = 0.04$, $w_{EI} = 0.5$, $w_{ff} = 0.8$, $I_0 = 1.496$; $\tau_r = 10$ ms, $\tau_I = 1$ ms, $\tau_{ff} = 15$ ms, $\tau_N = 150$ ms. The implementation differs from Wong and Wang (2006) in three respects: the transfer function is a saturating sigmoid, competition between the pools goes through an explicit inhibitory population, and a lagging feed-forward inhibition is added. Parameter values could therefore not be taken directly from their study. We chose them by hand so that each pool is bistable, with a low state near 0 Hz and a high state of about 85 Hz, and so that shared inhibition allows only one pool to be active at a time. The cue is a 50 ms input of amplitude 1 to the cued pool, after 300 ms of baseline, followed by a 1,600 ms delay. Equations were integrated with Euler steps of 0.1 ms. The memory was scored correct when, in the last 100 ms, the NMDA gating of the cued pool exceeded the uncued pool's by more than 0.05. We scored NMDA gating rather than rates because the model may be silent at the end of the delay, and a strong attentional input, modelling the response period, restarts the pool with more NMDA gating and so recovers the correct response (not shown).

Our main modification to this model is the common input $m(t)$ to both pools, which represents fluctuating attention. It is 1 except during dips, where it drops to $1 - \boldsymbol{\delta}$ with 10 ms ramps. Dip lengths are gamma-distributed (mean $\boldsymbol{\mu_d}$, shape $\boldsymbol{k}$), and the gaps between dips are exponential with mean $1/\boldsymbol{\lambda_d}$.

STP model

To compare our model with a short-term plasticity account, we replaced the NMDA gating s_i in the recurrent input by a saturating synaptic activation without a time constant of its own, $S_i = g_i/(1 + g_i)$, with $g_i = \beta r_i u_i / U$, where u_i is Tsodyks–Markram short-term facilitation (Mongillo et al., 2008; Mi et al., 2017):

$$\tau_f \frac{du_i}{dt} = U - u_i + \tau_f U(1 - u_i) r_i,$$

with $U = 0.3$ and $\tau_f = 1.5$ s. To simplify fitting and to give the STP model the best chance to fit the data, synaptic depression was switched off: with depression ($\tau_d = 0.3$ s), each re-ignition ended too early and *On states* were too short (112 vs 172 ms in the data). All other parameters were as in the NMDA model, and the STP model was fitted in the same way to the decoder using all neurons, with an additional penalty for population spikes before the cue. Best fit: $\boldsymbol{w_s} = 0.69$, $\boldsymbol{\lambda_d} = 5.5$/s, $\boldsymbol{\mu_d} = 212$ ms, $\boldsymbol{k} = 0.70$, $\boldsymbol{\delta} = 0.39$, $\boldsymbol{\sigma} = 0.005$. Both models matched the *On/Off* statistics about equally well (loss 9.7 for the NMDA model and 10.4 for the STP model, on 2,112 trials) and held the memory in nearly all trials.

Spiking network

We used the spatial working-memory network of Compte et al. (2000) with minimal changes. It consists of 2,048 pyramidal neurons and 512 interneurons, modelled as leaky integrate-and-fire neurons and arranged on a ring by preferred location. Recurrent excitation is mediated by NMDA receptors with a voltage-dependent Mg^{2+} block and saturating gating. Connections between pyramidal neurons decrease with the difference in their preferred locations (Gaussian footprint, $\sigma = 18°$), and all connections involving interneurons are uniform. All neurons receive external excitatory Poisson input through AMPA receptors. Neuronal and synaptic parameters were those of Compte et al. (2000), with four changes. First, the NMDA decay time was 150 ms instead of 100 ms, as in the rate model, and the NMDA conductance between pyramidal neurons was lowered from 0.381 to 0.33 nS so that the network remained bistable with the slower decay. Second, we made the neurons heterogeneous, so that their activity would resemble recorded neurons more closely. Firing thresholds varied across neurons (SD 0.3 mV), and each pyramidal neuron received a random fraction of its recurrent excitation through a broad, weakly tuned footprint instead of the narrow one. To keep the bump stable despite this weaker tuned input, the peak of the narrow footprint was raised from 1.62 to 3.0. Third, the cue was a 50 ms current to the pyramidal neurons around the cued location, as long as in the task and stronger than in Compte et al. (2000) (500 pA), so that it reliably started a bump. Fourth, as in the rate model, the external input to all pyramidal neurons was scaled by a common attentional signal $m(t)$ that dropped at random times and briefly overshot when it recovered. Equations were integrated with Euler steps of 0.1 ms.

As in Compte et al. (2000), the membrane potential V of each neuron obeys

$$C_m \frac{dV}{dt} = -g_L(V - E_L) - I_{\text{ext}} - I_{\text{NMDA}} - I_{\text{GABA}} + I_{\text{cue}},$$

with $C_m = 0.5$ nF and $g_L = 25$ nS for pyramidal neurons, $C_m = 0.2$ nF and $g_L = 20$ nS for interneurons, $E_L = -70$ mV, a firing threshold of -50 mV, a reset potential of -60 mV and refractory periods of 2 ms (pyramidal neurons) and 1 ms (interneurons). The synaptic currents are

$$\begin{aligned} I_{\text{ext}} &= g_{\text{ext}}\, s_{\text{AMPA}}(V - E_E), \\ I_{\text{NMDA}} &= g_{\text{NMDA}}\, B(V)(V - E_E) \textstyle\sum_j W_{ij}\, s_{\text{NMDA},j}, \\ I_{\text{GABA}} &= g_{\text{GABA}}(V - E_I) \textstyle\sum_j s_{\text{GABA},j}, \end{aligned}$$

with $E_E = 0$ mV, $E_I = -70$ mV and $B(V) = 1/(1 + [\text{Mg}^{2+}] \exp(-0.062\,V)/3.57)$, $[\text{Mg}^{2+}] = 1$ mM. The AMPA and GABA_A gating variables increase by 1 at each presynaptic spike and decay with $\tau_{\text{AMPA}} = 2$ ms and $\tau_{\text{GABA}} = 10$ ms. The NMDA gating of each pyramidal neuron obeys

$$\frac{dx}{dt} = -\frac{x}{\tau_x} + \sum_k \delta(t - t_k), \qquad \frac{ds_{\text{NMDA}}}{dt} = -\frac{s_{\text{NMDA}}}{\tau_{\text{NMDA}}} + \alpha\, x(1 - s_{\text{NMDA}}),$$

with $\tau_x = 2$ ms, $\alpha = 0.5$ ms^{-1} and $\tau_{\text{NMDA}} = 150$ ms. External Poisson input arrives at 1.8 kHz, with $g_{\text{ext}} = 3.1$ nS onto pyramidal neurons and 2.38 nS onto interneurons. Recurrent

conductances are $g_{\mathrm{NMDA}} = 0.33$ nS between pyramidal neurons and 0.292 nS from pyramidal neurons to interneurons, and $g_{\mathrm{GABA}} = 1.336$ nS onto pyramidal neurons and 1.024 nS onto interneurons. Between pyramidal neurons with preferred locations θ_i and θ_j,

$$W_{ij} = J^- + (J^+ - J^-) \exp\left(-\frac{(\theta_i - \theta_j)^2}{2\sigma^2}\right),$$

normalised to a mean of 1, with $\sigma = 18^\circ$ and $J^+ = 3.0$. For heterogeneous tuning, pyramidal neuron i received a fraction h_i of this input (h_i uniform between 0 and 1) through $1 + 0.25\cos(\theta_i - \theta_j)$ instead of W_{ij}. The external input rate to pyramidal neurons was multiplied by $m_{\mathrm{eff}}(t) = m(t) + A\,[m(t) - \bar{m}(t)]$, where $m(t)$ is the attentional signal, $\bar{m}(t)$ follows $m(t)$ down into each dip and recovers with a time constant of 150 ms, and $A = 1$.

The dip statistics of the spiking network were set by hand rather than fitted, but differ only slightly from those of the fitted rate model. Dips started at random times (Poisson, 7 per s instead of 10.2 per s), lasted longer (gamma distribution with the same shape, mean 150 ms instead of 90 ms) and were shallower (50% instead of 60% of the drive), with 10 ms ramps as in the rate model. The spiking network needed longer dips because its decoding was stronger. With 90 ms dips, the 100 ms decoding windows bridged most silences and *Off states* were rare. In both models, the drive briefly overshoots when the input recovers. In the rate model, this overshoot comes from the lagging feed-forward inhibition ($\tau_{ff} = 15$ ms). In the spiking network, a fast overshoot made all pyramidal neurons fire together, which saturated their NMDA gating and erased the memory, so the overshoot decayed with a time constant of 150 ms.

Sessions of 480 trials (60 per cue location, 8 locations) were simulated, each with 500 ms of fixation, a 50 ms cue and a 1,500 ms delay with dips throughout. In each session, 330 neurons (264 pyramidal neurons and 66 interneurons) were sampled at random across the ring and read out exactly as the recordings, with the same windows, decoder, label-shuffle null and *On*/*Off* labelling (4 sessions, Fig. 4b). Memory tuning in *On* and *Off states* was computed as in Panichello et al. (2024). Firing rates in 100 ms windows were z-scored across trials, the recorded neurons were split at random into two halves, and *On* and *Off states* were labelled with the decoder of one half. The delay tuning of each held-out neuron was computed separately for *On* and *Off* time points and aligned to its preferred cue, and the same analysis with shuffled cue labels was subtracted to remove the bias of this alignment (10 random splits, 330 neurons of one session).

Delay activity during *On* and *Off states* (Fig. 4d, e) was analysed in the same way in the recordings and in the network. In each session, neurons were split at random into two halves. *On* and *Off states* were labelled from one half, with the decoder and criteria above, and all other analyses used the other half. In the recordings, states were labelled from the 20 most cue-selective units of the labelling half. In the network, they were labelled from all of its neurons, because its 20 most selective neurons often preferred the same cue and were silent on trials of the other cue while the bump was still active. Delay activity (z-scored firing rates in 100 ms windows, 500–1,400 ms after cue onset) was averaged for each of the eight cue locations, and PCA was applied to these condition means. Axes were estimated on one half of the trials, used to project the other half, and aligned across halves by an orthogonal Procrustes rotation. Each trial's position was averaged over its *On* and, separately, its *Off* time points. Its alignment was the cosine between this position and the direction of its cue's mean in the other half of the trials, minus the same quantity with cue labels shuffled within each half (50 shuffles). Activity before the cue, projected on the same axes, served as a null (window centres from −400 ms to cue onset in the recordings and to −50 ms in the network, whose neurons respond to the cue within a few milliseconds). Ten simulated sessions were analysed.

Data

We reanalysed the recordings of Panichello et al. (2024): 25 sessions, correct trials only (error trials were used only in the behavioural analysis), all cue locations (13,190 trials; 329 units per

session on average). For each time bin, an L2-regularised logistic decoder ($\lambda = 10{,}000$) was evaluated with approximate leave-one-out and z-scored against 30 shuffled label sets per cue pair. Decoded populations were all neurons, the k most cue-selective neurons (ranked by split-half ANOVA), and k randomly chosen neurons (10 draws, averaged within sessions), with $k = 5$, 20 and 50. Decoded populations of the same size were compared within sessions with two-sided Wilcoxon signed-rank tests on the session values (random draws averaged within each session).

Behavioural analysis

To relate *On/Off states* to behaviour, we also labelled error trials (15,468 trials in total, 2,278 errors). Error trials were decoded with decoders fitted on all correct trials of the same cue pair, so they were also out of sample, and their rates were z-scored with the mean and SD of correct trials. For each trial, we computed the total *Off* time and the total *On* time in the delay (500–1,400 ms after cue onset), the number of *Off states* and their mean length (13,962 trials with at least one *Off state*). For each statistic, the probability of an error was modelled with a logistic mixed model, error $\sim$ z(stat) + t + t^2 + monkey + (1 + z(stat) || session), where z(stat) is the statistic z-scored within each session, so that only within-session variation is used, t is the time within the session, and each session has a random intercept and an uncorrelated random slope. Monkey was a fixed effect, as three monkeys are too few to estimate a random effect. Models were fitted with the Laplace approximation and tested with likelihood-ratio tests against the same model without the statistic. For the fit of τ_N, trials were pooled over sessions and binned by *Off* time in 90 ms bins (one tenth of the delay window).

Decoding read-out, *On/Off* labelling and statistics

To compare the model with the recordings, we measured *On* and *Off states* as close as possible to how they were measured in the data (Panichello et al., 2024), where they are defined on the z score of a decoder trained on the neurons recorded in a session. The model does not generate the activity of individual neurons. Instead, it describes the latent dynamics of a working-memory circuit, from which some neurons were recorded. Labelling *On/Off states* from the pool rates of the model directly would ignore the read-out noise inherent to recording only a small sample of the neurons in the circuit. We therefore converted the model's activity into a decoding z score with the same time resolution, noise and labelling as the data (Supplementary Fig. 1).

At each time step, the decoding evidence was the difference between the cued and uncued pool rates, $\Delta r(t)$. Δr is large when the cued pool is active and near zero when both pools are silent, and it was averaged in 100 ms windows stepped by 10 ms, as in Panichello et al. (2024). We modelled the z score as $z(t) = c\,\Delta r(t) + \eta(t)$, where the gain c converts rates into z units and η is the read-out noise, the variability of a decoder trained on a finite number of neurons and trials (Supplementary Fig. 1a).

The noise η dynamics was estimated from the data before the cue, where the decoder z carries no memory signal and therefore reflects read-out noise alone. The standard deviation of pre-cue z, averaged over the 25 sessions, was 1.06 for each decoded population, close to the value of 1 expected for a z score, so we gave η unit variance. In Panichello et al. (2024), z was computed in 100 ms windows stepped by 10 ms, so consecutive points share most of their spikes and the noise at two points Δt apart is correlated. We generated this part of η as white noise averaged in the same overlapping windows, $\varepsilon(t) = \frac{1}{\sqrt{10}} \sum_{j=0}^{9} w(t - 10j \text{ ms})$, where w are independent standard normal values drawn every 10 ms. In the data, z remained correlated beyond window size (100 ms, Supplementary Fig. 1b), possibly because the decoder output also drifts slowly from trial to trial. In the model, we captured this drift by adding to each trial a constant offset ξ, drawn independently for each trial from a standard normal distribution. The weight of this offset was set by ρ, the share of the noise variance that is constant within a trial, which we measured in the data as the correlation of pre-cue z between time points 200 ms apart. These points are

far enough apart that their windows do not overlap, so their correlation can only come from the per-trial drift ($\rho = 0.35$ for all neurons, 0.09 for the most selective neurons and 0.11 for random neurons). Thus $\eta(t) = \sqrt{1-\rho}\,\varepsilon(t) + \sqrt{\rho}\,\xi$, where ε is the window-averaged white noise. Both ε and ξ have unit variance, so η does too, and its autocorrelation is $(1-\rho)(1-\Delta t/100\text{ ms}) + \rho$ for $\Delta t < 100$ ms and ρ beyond, as in the data (Supplementary Fig. 1b).

We also calibrated the gain c on a separate property of the data, the level that z reaches when the memory is read out, which we call the decoding strength. The mean of the delay z would not serve this purpose, because the delay z mixes time points with strong and weak decoding (e.g. *On* vs *Off*). We therefore fitted a two-component Gaussian mixture to the delay z values (500–1,400 ms after cue onset) and took the mean of the upper component, which reflects the level of z during strong decoding (*On*) and depends much less on how often z is low (*Off*). For each decoded population, c was set so that the decoding strength of the model's delay z equaled the data's, averaged over sessions (Supplementary Fig. 1c,d).

States were labelled exactly as in Panichello et al. (2024), so that model and data are compared with identical definitions. *On states* are runs of $z > 1.96$ whose summed z exceeds the 95th percentile of the same quantity in pure noise η. *Off states* are runs of at least 5 points (50 ms) with $z < 0.84$. States were counted from 0 to 1,400 ms after cue onset, and fractions of the delay spent *On* or *Off* from 500 to 1,400 ms. Model statistics were computed from 2,112 simulated trials (Fig. 2). Data statistics are means over the 25 sessions; for random neurons, the 10 draws were first averaged within each session (Fig. 2).

Our read-out approximates a real decoder by a simple formula ($z(t) = c\,\Delta r(t) + \eta(t)$), so we tested whether the calibration recovers the gain and the *On/Off* statistics of a real decoder in synthetic data where the ground truth is known (Supplementary Fig. 1e–i). We used the fitted model to generate spikes and decoded them with the same data pipeline. The model was simulated for two cue locations, A and B. When A was cued, pool A was the cued pool and pool B the uncued one, and the reverse when B was cued, which gives the decoder two classes to discriminate, as in the data. The pool rates drove 330 Poisson neurons, the average number of units recorded per session (329). 30% of them followed pool A or pool B (half each) and the rest were untuned. Baseline rates were lognormal with a median of 5 Hz, and all neurons shared a per-trial gain, which produces slow trial-to-trial drift as in the data. We simulated 70 trials per cue location, close to the number of correct trials per location in a session, and 8 synthetic sessions at each of three tuning strengths, chosen so that the decoder's decoding strength matched that of the data's random-20, all-unit and top-20 decoders (Supplementary Fig. 1f). The fraction of tuned neurons matters only through this decoding strength, which the tuning strength sets. We then applied to the synthetic decoder output exactly the calibration used for the data. The recovered gain c grew in proportion to the tuning strength of the synthetic neurons, and its values (0.030, 0.050 and 0.086) were close to the gains of the model calibrated on the corresponding data decoders (0.029, 0.051 and 0.088; Supplementary Fig. 1g). With this gain and with ρ estimated from the pre-cue autocorrelation, the read-out z correlated with the decoder's as closely as two runs of the decoder on independent spikes from the same rates (Supplementary Fig. 1h). It reproduced the number and fraction of *Off states*, and undercounted *On states* by about one per trial (Supplementary Fig. 1i).

Fitting

Six parameters were fitted with the covariance matrix adaptation evolution strategy (Hansen and Ostermeier, 2001) to the decoder using all neurons: $\boldsymbol{w_s}$, $\boldsymbol{\lambda_d}$, $\boldsymbol{\mu_d}$, $\boldsymbol{k}$, $\boldsymbol{\delta}$ and $\boldsymbol{\sigma}$. The NMDA decay time was not fitted but fixed at $\tau_N = 150$ ms, within the range measured in cortex (Vicini et al., 1998; Wang et al., 2008, 2013). The loss combined the squared scaled errors of six statistics (*On and Off states* per trial, mean lengths and fractions of the delay), the squared Hellinger distance between model and data length distributions (states ≥ 40 ms), and a penalty when the memory was correct in fewer than 98% of trials. Best fit: $\boldsymbol{w_s} = 1.30$, $\boldsymbol{\lambda_d} = 10.2$/s, $\boldsymbol{\mu_d} = 90$ ms, $\boldsymbol{k} = 8.7$,

$\boldsymbol{\delta} = 0.60$, $\boldsymbol{\sigma} = 0.006$.

Simulations of memory against *Off* time

To compare with the behaviour, each fitted model was then simulated with random dips drawn from its fitted dip statistics (100,000 trials per model); the memory was held in 98.5% (NMDA model) and 99.99% (STP model) of these trials. To measure how long each model can hold the memory through a single silence (single-dip test, Fig. 3a), we simulated trials with only one dip of the common input, starting 500 ms after cue onset, with the model's fitted depth and a length from 0 to 1,000 ms in 25 ms steps (300 trials per length, 12,300 trials per model). Fig. 3a shows the fraction of these trials with the memory held against dip duration. Trials were read out with the all-units gain and noise of the fitted model, and the memory was scored as in the NMDA model, using the facilitation u_i instead of the NMDA gating in the STP model. *Off* time was computed as in the data.

NMDA time constant fitted to performance

The fitted NMDA model, with random dips drawn from the fitted dip statistics (the same dips for every τ_N) and the all-units read-out, was simulated with τ_N from 20 to 1,500 ms (17 values, 30,000 trials each) and all other parameters unchanged, so the *On/Off* statistics drift away from their fitted values as τ_N changes. For each τ_N, the probability that the memory was held was computed in the ten *Off*-time bins and interpolated in log τ_N. The model was linked to behaviour by P(correct) = p_0 × [P(held) + G × P(lost)], where G = 1/8 is the chance of a correct guess among the 8 cue locations after the memory is lost, and p_0 is the performance when the memory is held, which absorbs errors that do not depend on *Off* time. τ_N and p_0 were fitted by maximum likelihood to all 15,468 trials pooled (best $p_0 = 0.887$). Confidence intervals came from 1,000 bootstraps over sessions (25 sessions drawn with replacement, their trials pooled and refitted). The estimate depended little on G: the best τ_N was 129, 118 and 102 ms for G = 0, 1/4 and 1/2.

AI methodology

Claude Opus 5.5 was used to some extent for all aspects of this research, including assisting in writing text and code. I had 4 agents running in parallel, namely the mean-field model agent, the spiking network agent, the data analysis agent and paper writing agent. All agents interacted through github and pushed to different branches of the same repository. The paper writing agent had access to a google docs manuscript, which was proof read and edited manually. Conversion to latex code was done by the paper agent and latex code was not proofread by the author. Before publishing the code in github, an extra agent was guided to refactor the code generated by all agents and improve readability. Core modules (e.g. not plotting) were then proofread by the author. The author was guiding the work of 5 agents with similar strategies as when advising trainees while they are doing research (e.g. running controls, testing predictions, etc). In stark contrast to that analogy, everything published is the sole responsibility of the author.

References

Amit, D. J. and Brunel, N. (1997). Model of global spontaneous activity and local structured activity during delay periods in the cerebral cortex. *Cerebral Cortex*, 7:237–252.

Awh, E. and Jonides, J. (2001). Overlapping mechanisms of attention and spatial working memory. *Trends in Cognitive Sciences*, 5:119–126.

Barbosa, J., Stein, H., Martinez, R. L., Galan-Gadea, A., Li, S., Dalmau, J., Adam, K. C. S., Valls-Solé, J., Constantinidis, C., and Compte, A. (2020). Interplay between persistent activity and activity-silent dynamics in the prefrontal cortex underlies serial biases in working memory. *Nature Neuroscience*, 23:1016–1024.

Busch, N. A., Dubois, J., and VanRullen, R. (2009). The phase of ongoing EEG oscillations predicts visual perception. *Journal of Neuroscience*, 29:7869–7876.

Busch, N. A. and VanRullen, R. (2010). Spontaneous EEG oscillations reveal periodic sampling of visual attention. *Proceedings of the National Academy of Sciences USA*, 107:16048–16053.

Compte, A., Brunel, N., Goldman-Rakic, P. S., and Wang, X.-J. (2000). Synaptic mechanisms and network dynamics underlying spatial working memory in a cortical network model. *Cerebral Cortex*, 10:910–923.

Engel, T. A., Steinmetz, N. A., Gieselmann, M. A., Thiele, A., Moore, T., and Boahen, K. (2016). Selective modulation of cortical state during spatial attention. *Science*, 354:1140–1144.

Fiebelkorn, I. C. and Kastner, S. (2019). A rhythmic theory of attention. *Trends in Cognitive Sciences*, 23:87–101.

Fiebelkorn, I. C., Pinsk, M. A., and Kastner, S. (2018). A dynamic interplay within the frontoparietal network underlies rhythmic spatial attention. *Neuron*, 99:842–853.

Hansen, N. and Ostermeier, A. (2001). Completely derandomized self-adaptation in evolution strategies. *Evolutionary Computation*, 9:159–195.

Helfrich, R. F., Fiebelkorn, I. C., Szczepanski, S. M., Lin, J. J., Parvizi, J., Knight, R. T., and Kastner, S. (2018). Neural mechanisms of sustained attention are rhythmic. *Neuron*, 99:854–865.

Hempel, C. M., Hartman, K. H., Wang, X.-J., Turrigiano, G. G., and Nelson, S. B. (2000). Multiple forms of short-term plasticity at excitatory synapses in rat medial prefrontal cortex. *Journal of Neurophysiology*, 83:3031–3041.

Kiyonaga, A., Scimeca, J. M., Bliss, D. P., and Whitney, D. (2017). Serial dependence across perception, attention, and memory. *Trends in Cognitive Sciences*, 21:493–497.

Landau, A. N. and Fries, P. (2012). Attention samples stimuli rhythmically. *Current Biology*, 22:1000–1004.

Lundqvist, M., Herman, P., and Miller, E. K. (2018). Working memory: delay activity, yes! Persistent activity? Maybe not. *Journal of Neuroscience*, 38:7013–7019.

Lundqvist, M., Rose, J., Herman, P., Brincat, S. L., Buschman, T. J., and Miller, E. K. (2016). Gamma and beta bursts underlie working memory. *Neuron*, 90:152–164.

Markram, H., Wang, Y., and Tsodyks, M. (1998). Differential signaling via the same axon of neocortical pyramidal neurons. *Proceedings of the National Academy of Sciences USA*, 95:5323–5328.

Mi, Y., Katkov, M., and Tsodyks, M. (2017). Synaptic correlates of working memory capacity. *Neuron*, 93:323–330.

Miller, E. K., Lundqvist, M., and Bastos, A. M. (2018). Working memory 2.0. *Neuron*, 100:463–475.

Mongillo, G., Barak, O., and Tsodyks, M. (2008). Synaptic theory of working memory. *Science*, 319:1543–1546.

Mongillo, G. and Tsodyks, M. (2025). Synaptic encoding of time in working memory. *eLife*, 14:RP107005.

Mozumder, R., Wang, Z., Dang, W., Zhu, J., Hammond, B. M., Machado, A., and Constantinidis, C. (2026). Asynchronous firing and off states in working memory maintenance. *Cell Reports*, 45:116764.

Panichello, M. F. and Buschman, T. J. (2021). Shared mechanisms underlie the control of working memory and attention. *Nature*, 592:601–605.

Panichello, M. F., Jonikaitis, D., Oh, Y. J., Zhu, S., Trepka, E. B., and Moore, T. (2024). Intermittent rate coding and cue-specific ensembles support working memory. *Nature*, 636:422–429.

Pascucci, D., Tanrikulu, Ö. D., Ozkirli, A., Houborg, C., Ceylan, G., Zerr, P., Rafiei, M., and Kristjánsson, Á. (2023). Serial dependence in visual perception: a review. *Journal of Vision*, 23(1):9.

Rose, N. S., LaRocque, J. J., Riggall, A. C., Gosseries, O., Starrett, M. J., Meyering, E. E., and Postle, B. R. (2016). Reactivation of latent working memories with transcranial magnetic stimulation. *Science*, 354:1136–1139.

Stokes, M. G. (2015). 'Activity-silent' working memory in prefrontal cortex: a dynamic coding framework. *Trends in Cognitive Sciences*, 19:394–405.

Tao, W. and Libedinsky, C. (2024). Evidence of activity-silent working memory in prefrontal cortex. *bioRxiv*. doi:10.1101/2024.06.03.597259.

Thomson, A. M. (1997). Activity-dependent properties of synaptic transmission at two classes of connections made by rat neocortical pyramidal axons in vitro. *Journal of Physiology*, 502:131–147.

Urai, A. E., Doiron, B., Leifer, A. M., and Churchland, A. K. (2022). Large-scale neural recordings call for new insights to link brain and behavior. *Nature Neuroscience*, 25:11–19.

Vicini, S., Wang, J. F., Li, J. H., Zhu, W. J., Wang, Y. H., Luo, J. H., Wolfe, B. B., and Grayson, D. R. (1998). Functional and pharmacological differences between recombinant N-methyl-D-aspartate receptors. *Journal of Neurophysiology*, 79:555–566.

Wang, H.-X., Stradtman, G. G., Wang, X.-J., and Gao, W.-J. (2008). A specialized NMDA receptor function in layer 5 recurrent microcircuitry of the adult rat prefrontal cortex. *Proceedings of the National Academy of Sciences USA*, 105:16791–16796.

Wang, M., Yang, Y., Wang, C.-J., Gamo, N. J., Jin, L. E., Mazer, J. A., Morrison, J. H., Wang, X.-J., and Arnsten, A. F. T. (2013). NMDA receptors subserve persistent neuronal firing during working memory in dorsolateral prefrontal cortex. *Neuron*, 77:736–749.

Wang, X.-J. (1999). Synaptic basis of cortical persistent activity: the importance of NMDA receptors to working memory. *Journal of Neuroscience*, 19:9587–9603.

Wang, X.-J. (2001). Synaptic reverberation underlying mnemonic persistent activity. *Trends in Neurosciences*, 24:455–463.

Wang, X.-J. (2002). Probabilistic decision making by slow reverberation in cortical circuits. *Neuron*, 36:955–968.

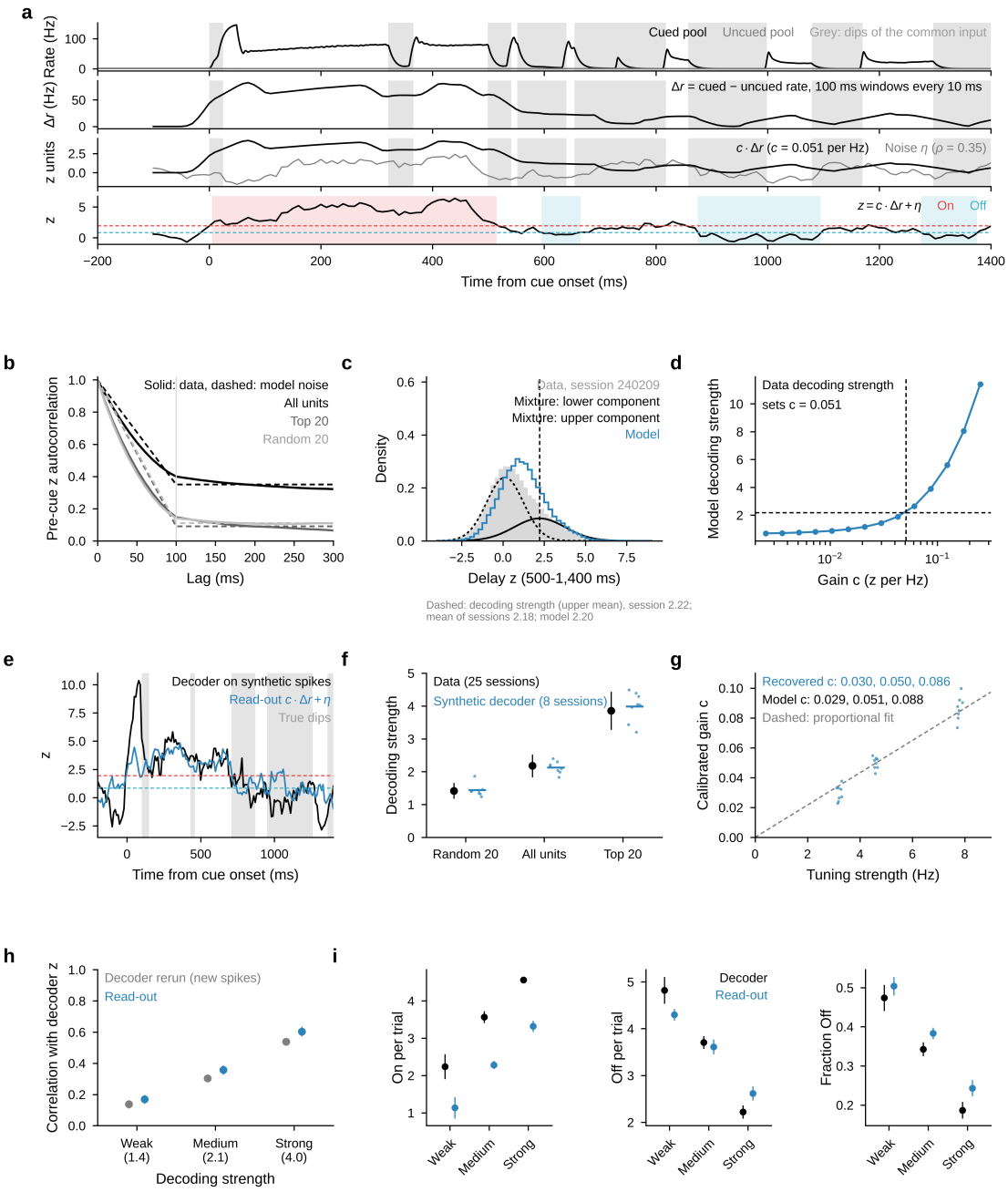
Wang, Y., Markram, H., Goodman, P. H., Berger, T. K., Ma, J., and Goldman-Rakic, P. S. (2006). Heterogeneity in the pyramidal network of the medial prefrontal cortex. *Nature Neuroscience*, 9:534–542.

Williams, A. H. and Linderman, S. W. (2021). Statistical neuroscience in the single trial limit. *Current Opinion in Neurobiology*, 70:193–205.

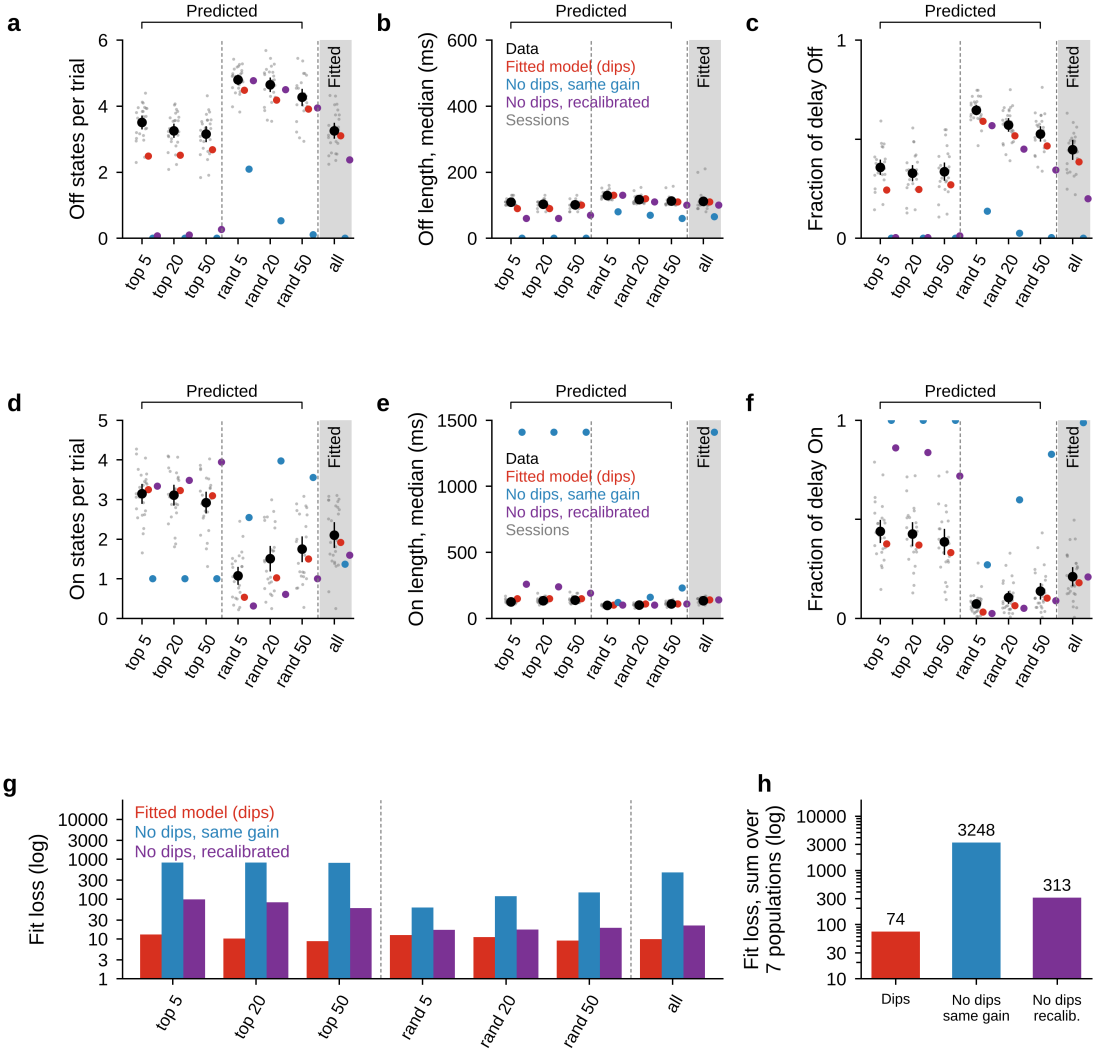
Wolff, M. J., Jochim, J., Akyürek, E. G., and Stokes, M. G. (2017). Dynamic hidden states underlying working-memory-guided behavior. *Nature Neuroscience*, 20:864–871.

Wong, K.-F. and Wang, X.-J. (2006). A recurrent network mechanism of time integration in perceptual decisions. *Journal of Neuroscience*, 26:1314–1328.

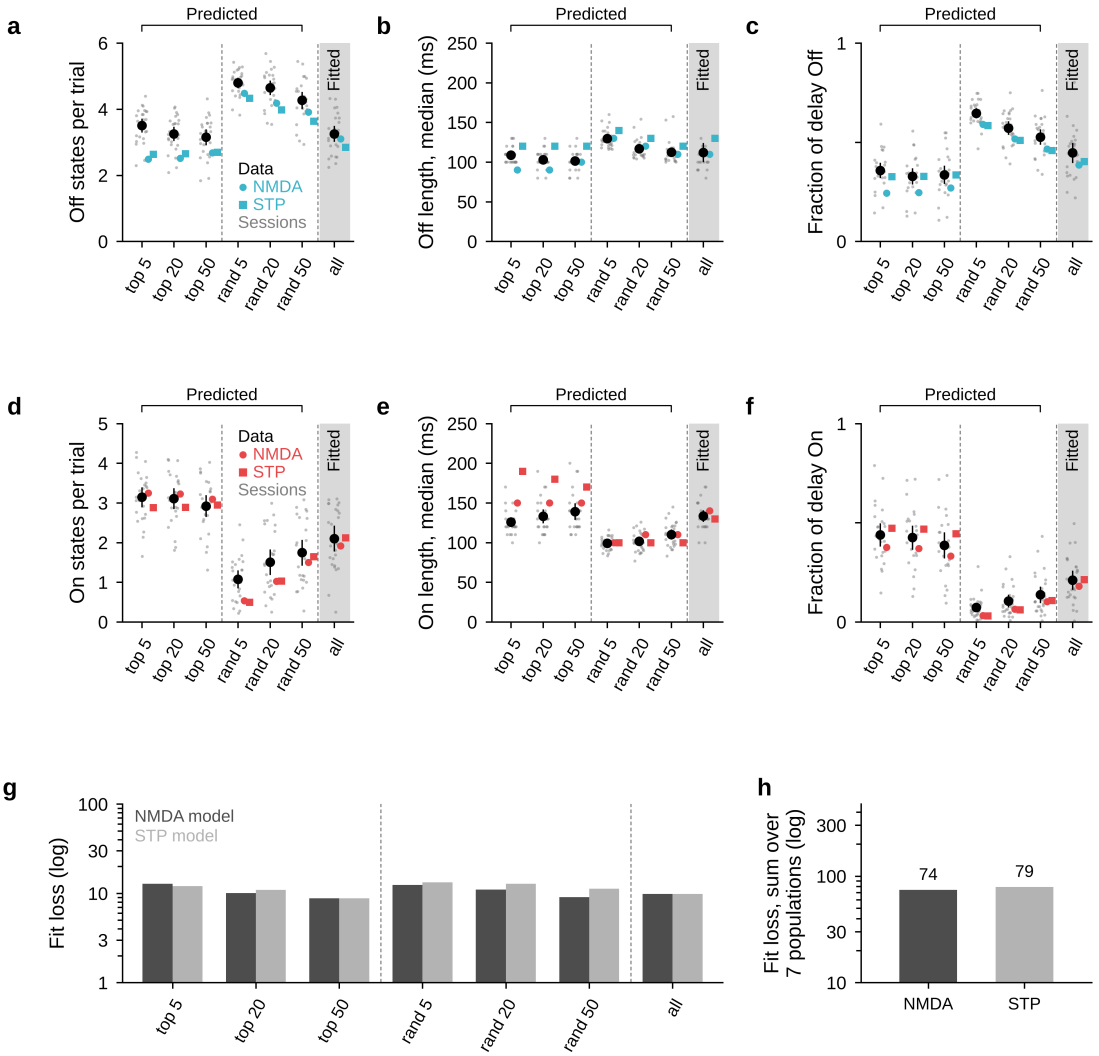
Supplementary Figures



Supplementary Figure 1: Reading the model out like the data. **a**, One model trial through the read-out. From top to bottom, rates of the cued (black) and uncued (grey) pools, with dips of the common input in grey; Δr, their difference averaged in 100 ms windows every 10 ms; $c \cdot \Delta r$ (black) and the read-out noise η (grey); and $z = c \cdot \Delta r + \eta$, with *On* (red) and *Off* (cyan) states labelled with the thresholds of Panichello et al. (2024) (dashed, z = 1.96 and 0.84). **b**, Autocorrelation of the decoder z before the cue, where there is no memory signal, for three decoded populations (solid; black, all units; dark grey, top 20; light grey, random 20), and of the model's noise η with the ρ of each population (dashed). The vertical line marks the 100 ms window, beyond which the correlation reflects the per-trial offset. **c**, Delay z (500–1,400 ms after cue onset) of one example session (grey; session 240209, whose decoding strength is closest to the mean over sessions), with the two components of a Gaussian mixture fitted to it (dotted, lower; solid, upper), and the model's delay z (blue). Dashed, decoding strength, the mean of the upper component (2.22 for this session, 2.18 averaged over sessions, 2.20 for the model). **d**, Decoding strength of the model as a function of the gain c. The data's decoding strength (horizontal line) sets c (vertical line). **e**–**i**, Validation on synthetic data. The fitted model drove 330 Poisson neurons, at three tuning strengths, and their spikes were decoded with the data pipeline. **e**, Example trial with the true dips (grey), the decoder's z (black) and the read-out $z = c \cdot \Delta r + \eta$ (blue). **f**, Decoding strength of the synthetic decoder (blue, 8 sessions per tuning strength) and of the data's random-20, all-unit and top-20 decoders (black, mean ± 2 SEM over 25 sessions). **g**, Gain c recovered by applying the data calibration to the synthetic decoder, against the tuning strength of the synthetic neurons. Dashed, proportional fit. **h**, Correlation of the read-out's delay z with the decoder's (blue), and of a rerun of the decoder on new spikes (grey), which sets the ceiling for two noisy read-outs. **i**, *On and Off* statistics of the decoder (black) and of the read-out (blue). **h**, **i**, mean ± 2 SEM over 8 synthetic sessions.



Supplementary Figure 2: Off states are not produced by the read-out noise alone. *On/Off* statistics of the seven decoded populations, as in Fig. 2: **a**, *Off states* per trial; **b**, median *Off* length; **c**, fraction of the delay *Off*; **d**, *On states* per trial; **e**, median *On* length; **f**, fraction of the delay *On*. Black circles, data (mean ± 2 SEM over 25 sessions). Grey dots, single sessions. Red, fitted model (with dips). Blue, the same model with a constant common input (no dips), read out with the identical read-out (same gain, noise and labelling) for each population. Purple, the no-dip model with the read-out gain recalibrated to each population's decoding strength in the data. Without dips, the identical read-out produced no *Off states* for the top-unit and all-unit decoders and few for random neurons. With the gain recalibrated, noise alone reproduced the *Off states* of the random populations but not those of the top-unit decoders (0.1–0.3 vs 3.2–3.5 *Off states* per trial; summed fit loss over the seven populations 313, vs 74 for the fitted model). g, Fit loss of each condition for each decoded population (the loss used for fitting, Methods; log scale). h, Fit loss summed over the seven decoded populations.



Supplementary Figure 3: The STP model fits the *On/Off* statistics about as well as the NMDA model. As Fig. 2, for the NMDA model (circles) and the STP model, in which short-term facilitation replaces NMDA gating (squares; Methods). Both models were fitted to the decoder using all neurons only (grey band), and the other six decoded populations are predictions obtained by changing only the read-out gain and noise. Black circles, data (mean ± 2 SEM over 25 sessions). Grey dots, single sessions. Cyan and red symbols, model *Off and On* statistics. g, Fit loss of each model for each decoded population (the loss used for fitting, Methods; log scale). h, Fit loss summed over the seven decoded populations.